PROJECT PROPOSAL

Automated Food Ordering and Booking System for Trust Fund Restaurant
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ABSTRACT

The current food ordering, booking, and meal tracking process at the campus cafeteria (Trust Fund Restaurant) is fully manual and inefficient. Students and staff place orders through phone calls or long queues, while all records are kept in hardcover notebooks. This leads to errors, slow service, unorganized order history, wasted food, poor support for vegetarians, and losses from untracked monthly meal subscribers.
This proposal presents an **Automated Food Ordering System** that includes real-time menus, digital booking, order tracking, online payments (Mpamba + Airtel Money), QR-based ticket collection, meal substitution, notifications, and data analytics. The goal is to modernize the cafeteria, improve customer satisfaction, reduce leftovers, and make operations faster and more reliable.

1. INTRODUCTION

The Trust Fund Restaurant serves many students, lecturers, and staff daily. However, the current ordering system is slow, heavily manual, and unreliable. Customers face long queues, delays, and a lack of information about available meals. Kitchen staff juggle phone calls, cooking tasks, and manual record keeping, which affects efficiency and accuracy.
To solve these challenges, this proposal introduces a **digital ordering and booking system** that automates menu management, order placement, notification alerts, monthly subscriber tracking, and meal collection using QR codes. The system aims to create a smoother, faster, and more organized food service experience.

1.1 PROJECT HISTORY

The project idea started after observing challenges at the cafeteria for several weeks. Students complained about long queues, poor communication, and surprise menu changes. Some vegetarian students returned without food because their meals were not properly prepared or prioritized. Staff struggled with phone orders, piles of notebooks, and pressure from customers. Interviews and real-life observations showed that a digital solution could significantly improve service delivery, food planning, record organization, and customer satisfaction.

1.2 PROBLEM DEFINITION

The current ordering and booking process at the Trust Fund Restaurant is fully manual, slow, and disorganized. After conducting interviews, observing operations, and reviewing customer experiences, several major problems were identified. These challenges affect both customers (students, lecturers, staff) and cafeteria workers, leading to delays, poor service quality, financial losses, and food wastage. The main problems are explained below:

I. Manual Order Placement Through Phone Calls

Customers place food orders by calling the kitchen directly. This creates confusion because kitchen staff are often busy cooking, serving, or cleaning. As a result, they may forget orders, write them incorrectly, or delay responding to calls. This slows down operations and leads to customer dissatisfaction.

II. Paper-Based Calculations and Record Keeping

All order records, payments, bookings, and daily activities are written in hardcover notebooks. This method is prone to human error, difficult to search, and easy to lose or damage. It also requires more time, especially during busy hours, which slows down service delivery.

III. Poorly Organized Order History

Because records are kept on paper, retrieving old orders for reference is almost impossible. There is no simple way to check past bookings, total daily sales, customer trends, or popular meals. This affects reporting, planning, and decision-making.

IV. Fully Manual Order Placement System

Everything—from taking orders to confirming them—is done manually. Staff must listen carefully, write details down, and process each customer one-by-one. This increases the chances of mistakes and slows down meal preparation.

V. Use of Hardcover Books for Bookings and Records

All booking information from departments, events, or individuals is stored in physical books. These books become full quickly, get torn, and make it difficult to trace booking history or verify past reservations. This leads to double bookings or disputes.

VI. Kitchen Staff Have Tight Schedules

Workers must cook, answer calls, take orders, update notebooks, and serve customers at the same time. This heavy workload causes stress, service delays, and frequent mistakes in order preparation.

VII. Frequent Leftovers

Since there is no digital system to track daily demand or popular foods, kitchen staff prepare meals based on assumptions. This often leads to cooking too much food, resulting in leftovers and financial losses. Other times, not enough food is prepared, leading to customer frustration.

VIII. Vegetarians Are Not Properly Served

This problem is severe and needs special attention. Vegetarians often return without food because:

- They do not know the menu in advance.
- Their meals are not prioritized during preparation.
- They do not pre-order because the process is manual.
- Staff may forget or confuse vegetarian orders when the kitchen is busy.

As a result, some vegetarian customers simply walk away without being served, wasting their time and negatively affecting inclusiveness.

IX. Many Students Do Not Know They Are Allowed to Order

Lack of communication creates misunderstandings. Some students assume that ordering is only for staff or certain departments, while in reality *any student* should be able to order. This reduces student participation and cafeteria revenue.

X. Manual Booking System Leads to Errors

Departments, clubs, and individuals book meals using paper forms or verbal agreements. This system leads to:

- Miscommunication
- Lost booking records
- Double bookings
- Delayed service

Long queues form when many people come at once to make bookings.

XI. Long Queues During Ordering and Booking

During peak hours, customers wait for long periods just to check availability or place an order. This wastes time, especially for lecturers with tight schedules or students who want to eat quickly between classes.

XII. No Real-Time Menu

The menu is created randomly based on what staff feel like cooking, instead of customer preferences or demand data. Customers do not know:

- What meals are available
- Which foods are finished
- What ingredients are being used

This leads to confusion and a poor dining experience.

XIII. Customers Do Not Know When Their Food Is Ready

Since everything is manual, there is no notification system to inform customers that their meal is ready. Many customers wait longer than necessary or return to the cafeteria multiple times to check.

XIV. Customers Waste Time and Money Traveling to the Cafeteria

Some customers drive or walk long distances to the cafeteria only to find that their preferred meal is already finished. This is especially frustrating for:

- Busy lecturers with limited time
- Students with classes
- Customers who spend money on transport

A digital system would prevent such wasted trips.

XV. Monthly Meal Subscribers Are Not Tracked Properly

Some students pay for a full month of food access. However, because there is no tracking:

- Students can claim meals they already received
- Other students can collect prepaid meals pretending they are the subscriber
- Staff cannot confirm the identity of the person collecting

This causes financial losses and unfair service.

Summary of the Problem

The Trust Fund Restaurant faces serious operational, organizational, and communication challenges due to manual systems. These issues slow down service, reduce customer satisfaction, and cause unnecessary costs. A digital solution is necessary to fix these problems and modernize the cafeteria.

1.3 OBJECTIVES / GOALS

Main Objective

The main objective of this project is to design and develop a fully automated food ordering and meal-booking system for the Trust Fund Restaurant that replaces all manual processes with a fast, digital, and efficient platform. The system aims to modernize cafeteria operations by introducing real-time menu updates, digital booking, online payments, QR-based order collection, automated notifications, and accurate record management. The overall goal is to improve service delivery, reduce errors, shorten queues, support special diets, prevent financial losses, and enhance customer satisfaction for students, lecturers, and staff.

Specific Objectives

- Digitize the food ordering and booking process.
- Provide a real-time menu with availability updates.
- Notify customers when their food is ready.
- Automate order history and monthly subscriber tracking.
- Reduce queues and long waiting times.
- Support vegetarians and special diets through prioritization.
- Reduce food waste by analyzing demand.
- Integrate digital payments (TNM Mpamba + Airtel Money).
- Ensure customers can change unavailable meals easily.

2. AREA UNDER STUDY

2.1 Scope

The Automated Food Ordering and Booking System will focus on improving how the Trust Fund Restaurant manages daily operations. The scope defines what the system will do and the boundaries of the project. This ensures that development is clear, organized, and aligned with the needs identified during research.
Below is the expanded explanation of each module included in the system:

1. **Digital Ordering (Web & Mobile)**

The system will allow students, lecturers, staff, and monthly subscribers to place food orders through an online platform. Users will be able to browse the menu, choose meals, select quantity, and complete payment using a mobile phone or computer. This will reduce phone calls and eliminate physical queues.

2. **Real-Time Menu Updates**

The menu will be updated instantly by cafeteria administrators. Customers will always see current meal availability, prices, ingredients, and portions left. This helps prevent disappointment when customers arrive and find meals finished.

3. **Meal Booking and Scheduling**

Departments, clubs, and individuals will be able to book meals in advance for meetings, events, workshops, or daily needs. They can choose the date, time, and quantity. The system will also support recurring bookings for monthly meal subscribers.

4. **Payment Integration (Mpamba + Airtel Money)**

The system will support cashless transactions through TNM Mpamba and Airtel Money. This will allow customers to complete their orders quickly and reduce money-handling errors. Payment confirmations will be automatic and recorded in the system.

5. **QR-Code Ticketing for Meal Collection**

Each order will automatically generate a secure QR code. Customers will scan the ticket at the cafeteria during food collection. This ensures that only the correct customer collects the order, and it prevents theft or confusion, especially for monthly subscribers.

6. **Order Tracking and Notification System**

Customers will receive real-time notifications such as:

- Order received
- Order accepted
- Order being prepared
- Order ready
- Order collected
  This reduces waiting times and prevents customers from making unnecessary trips to the cafeteria.
- **Meal Substitution Feature**

If a meal becomes unavailable (finished or not prepared), the system will prompt the customer to choose another meal within the same price range. This prevents frustration and ensures smooth order processing.

- **Admin Dashboard**

Administrators will manage menus, bookings, daily orders, payments, inventory indicators, and reports from one centralized dashboard. They can also monitor customer traffic, peak hours, and popular meals.

- **Kitchen Ticket Management**

Kitchen staff will receive orders instantly on a dedicated dashboard. They can track pending orders, preparing orders, ready orders, and collected orders. This replaces manual writing in notebooks and improves workflow.

- **Monthly Subscriber Tracking**

The system will record prepaid subscribers and track each meal collected. Each time the subscriber scans a QR code, the system deducts one meal from their remaining balance. This prevents meal theft and ensures accurate accounting.

- **Reports and Analytics for Decision-Making**

The system will generate daily, weekly, and monthly reports, including:

- Best-selling meals
- Food demand patterns
- Leftover trends
- Revenue reports
- Subscriber usage
- Peak ordering hours
  This helps management plan ingredients, reduce wastage, and improve service quality.

Boundary of the Scope

Only the food collection process (physically giving food to customers) will remain manual.
All data handling, communication, payment, tracking, and management will be fully automated.

2.2 Deliverables

The project will produce the following deliverables upon completion:

- **Fully Functional Automated Food Ordering System**

A responsive web and mobile system that customers and administrators can use for ordering, booking, payments, and management.

- **QR Ticket Generator**

A secure module that automatically creates QR codes for every order and monthly subscription meal.

- **Real-Time Menu Module**

A dynamic section where administrators update the menu, meal quantities, and availability instantly.

- **Admin & Kitchen Dashboards**

Two separate dashboards:

  - Admin dashboard for management, bookings, reports, and menu updates
  - Kitchen dashboard for order viewing, meal preparation tracking, and collection confirmation

- **Payment Integration Module**

A safe and reliable connection with TNM Mpamba and Airtel Money for digital payments and automated confirmations.

- **Monthly Subscriber Tracking System**

A feature that records prepaid meals, tracks daily collections, prevents double collection, and provides transparency for both staff and subscribers.

- **Reports and Analytics Engine**

Charts, summaries, and downloadable reports that help management make informed decisions.

- **Order Notification System**

A communication module that sends alerts to customers about order status changes.

3. PROPOSED SYSTEM FEATURES

The Automated Food Ordering and Booking System will include a wide range of modern and user-friendly features designed to solve the problems identified in the current manual process. These features focus on improving efficiency, reducing errors, supporting customers with special needs, and providing transparency throughout the ordering and meal-collection process.

3.1 Real-Time Digital Menu

The system will display a digital menu that updates instantly whenever the cafeteria staff make changes. Each meal will include details such as name, price, portion size, ingredients, and availability status (e.g., “Available,” “Almost Finished,” “Out of Stock”).
This ensures customers always know what meals are offered without physically visiting the cafeteria.

3.2 Smart Order Placement

Students, lecturers, and staff can place orders using a smartphone or computer. The interface will be simple and intuitive, allowing users to select meals, view prices, choose quantity, make payment, and receive order confirmation.
This eliminates the need for phone calls and reduces queues at the cafeteria.

3.3 Automatic Ticket Generation

After placing an order, the system automatically generates a unique QR-code ticket. Customers will show this ticket at the cafeteria, where staff will scan it to confirm collection.
This feature prevents meal theft, avoids confusion, and ensures accurate tracking for prepaid monthly subscribers.

3.4 Mobile Money Integration (Mpamba & Airtel Money)

The system will support secure and automated mobile money payments. Customers will pay directly through TNM Mpamba or Airtel Money, and the system will record transactions instantly.
This reduces cash handling, improves safety, and speeds up the ordering process.

3.5 Order Notifications

Customers will receive notifications at every important stage of the order.
These include:

- Order Accepted – The kitchen has confirmed the order.
- Preparing – The meal is being cooked or packed.
- Ready – The meal is ready for collection.

- Collected – The meal has been scanned and handed out.

This prevents unnecessary waiting and helps customers plan their time better.

3.6 Meal Substitution Feature

If a selected meal becomes unavailable due to high demand or ingredient shortage, the system will immediately notify the customer and allow them to choose another meal within the same price range.
This ensures customers do not lose their money or feel frustrated.

3.7 Special Diet Support (Vegetarian)

The system will include advanced dietary support through:

- Clear labeling of special meals
- Prioritized preparation for vegetarian and special-diet customers
- Pre-ordering options to ensure their meals are always available
  This ensures inclusiveness and reduces the number of vegetarian customers leaving without food.

3.8 Booking Module

Departments, clubs, and individuals can book meals digitally for meetings, workshops, or events. They can choose the date, time, number of people, and type of meal required.
This reduces paperwork and eliminates double bookings.

3.9 Queue Management & Estimated Time of Arrival (ETA)

When the cafeteria is busy, the system will show the expected preparation time for each order. Customers can see how long they need to wait before their food is ready.
This reduces congestion and helps users avoid unnecessary waiting at the counter.

3.10 Monthly Subscriber Tracking

Subscribers who pay for monthly meals will be registered in the system. Each meal they collect will be scanned using a QR code, deducting one meal from their remaining balance.
This prevents meal theft, fraud, and confusion regarding how many meals remain.

3.11 Analytics & Reports

The system will generate meaningful data for management, including:

- Most popular meals
- Daily, weekly, and monthly sales
- Leftover analysis
- Peak ordering hours
- Subscriber activity
  These reports help the restaurant improve planning, reduce food wastage, and understand customer preferences.

4. PROJECT PLAN

Phase 1: Requirements Gathering

• Interview cafeteria staff, kitchen workers, students, and lecturers
• Observe current ordering, booking, and meal-collection workflows
• Identify user needs, problems, and expectations

Phase 2: System Analysis

• Prepare use cases
• Develop Data Flow Diagrams (DFDs)
• Create the Entity Relationship Diagram (ERD)
• Define system inputs, outputs, and security needs

Phase 3: System Design

• Create UI/UX wireframes for customers, admin, and kitchen staff
• Design system architecture (frontend, backend, database)
• Create database schema and table relationships
• Plan QR-code workflow and payment integration

Phase 4: Development

• Develop backend APIs and frontend interfaces
• Integrate Mpamba and Airtel Money
• Implement QR-code generation and scanning
• Build ordering, booking, notifications, and analytics modules

Phase 5: Testing

- Unit testing for each feature
- Integration testing for combined modules
- User Acceptance Testing (UAT) with cafeteria staff

Phase 6: Deployment

- Deploy system to a hosting server
- Configure database and payment gateways
- Train cafeteria staff
- Provide final system documentation

Resources Needed

- Laptops
- Internet access
- Hosting platform
- Development tools and frameworks
- Database server

5. EXPECTED IMPACT

- Reduced queues by up to 50%
- Zero confusion on meal availability
- Accurate records for all orders
- Reduced leftovers by demand forecasting
- Better service for vegetarians and special diets
- Reduced staff workload
- No more monthly meal theft
- Improved customer satisfaction

6. CONCLUSION

The current manual food ordering and booking system at the Trust Fund Restaurant is inefficient, slow, and unreliable. With frequent errors, long queues, poor communication, and lack of transparency, customers and staff face daily challenges.
The proposed Automated Food Ordering System introduces a modern, reliable, and efficient solution that digitizes operations, improves accuracy, supports cashless payments, and enhances customer experience. This system is essential for improving service delivery, reducing financial losses, supporting special diets, and modernizing cafeteria operations.